Functional & Non-Functional Requirements

Target Audience

Our target audience consists of commuters and drivers in Singapore who are seeking reliable and convenient ways to plan their journeys. They value real-time information on carparks, traffic conditions, bus arrivals, and taxi availability, all accessible through a single platform. The system also caters to users who prefer simple, intuitive tools that save time and provide cost-effective travel recommendations.

Functional Requirements

1. Website shall provide account creation for first time users using verifiable information
   - 1.1. Account creation shall support two flows:
     - 1.1.1. OAuth-based sign-up using Google authentication.
     - 1.1.2. Email, Password and OTP sign-up
   - 1.2. Information entered shall be in the correct format
     - 1.2.1. OTP shall match the 6-digit code sent to the user's email, if any.
     - 1.2.2. Email shall be in the standard format.
     - 1.2.3. Password shall be at least 8 characters long,
       - 1.2.3.1. Password shall include at least one lower case
       - 1.2.3.2. Password shall include at least one upper case
       - 1.2.3.3. Password shall include either a special character or number.
   - 1.3. Website shall prohibit the creation of account if any of the information does not meet the requirements.
   - 1.4. Verification email shall be sent to the registered email address.

2. Website shall allow registered users to log in to their account
   - 2.1. When a user enters an incorrect email or password combination that does not tie to any registered account, the website shall prevent login and display an error message stating "Invalid email or Password. Try again."
   - 2.2. Website shall allow registered users to reset their password if they forget it
     - 2.2.1. Website shall send an email to registered users containing a link for them to reset their password.
   - 2.3. Website shall allow users to log out at any time after logging in.

3. Website shall request permission to access the user's location.
   - 3.1. If location is granted, zoom in to display the user's current location and their desired destination along with the relevant information on the map view.
     - 3.1.1. Website shall display nearby carparks (if any) with icons.

Functional & Non-Functional Requirements

3.1.2. Website shall display nearby and on-route roadworks (if any) with icons.

3.1.3. Website shall display the real-time locations of available moving taxis within a 1km radius of the user's current location using icons based on latest pulled API LTA Data.

3.1.4. Website shall display the nearby and on-route bus stops (if any) with icons.

3.1.5. Users shall be able to click on the carpark icon to view the following details

3.1.5.1. Name of the carpark

3.1.5.2. Number of available lots

3.1.5.3. Type of lots (e.g. multi-storey/surface/sheltered/basement)

3.1.5.4. Users shall be allowed to click refresh to get the latest number of available lots.

3.1.6. Users shall be able to click on the bus stop icon to view the following details

3.1.6.1. Bus stop name and code

3.1.6.2. Bus arrival timings

3.1.6.3. Bus diversions (if any)

3.1.6.4. Users shall be allowed to click refresh to get the latest bus arrival timings.

3.1.7. Users shall be able to click on traffic alert icons to view the following details

3.1.7.1. Type of alert (lane closure/roadwork/accident etc)

3.1.7.2. Affected roads

3.1.7.3. Alert descriptions

3.1.7.4. Users shall be allowed to click refresh to get the latest alerts or if traffic alert has been cleared.

3.1.8. Clicking on taxi icons shall not do anything. They are just there for visual density.

3.2. If location is not granted, the website shall show a zoomed-out map of Singapore.

3.2.1. When the user adjusts the map to focus on a specific road/district/neighbourhood, the website shall display traffic conditions, nearby carparks, taxi availability hotspots and bus stop locations as icons within a 2km radius.

3.2.2. Clicking on the carpark, bus stop and roadwork icons shall show the same information as in 3.1.5, 3.1.6 and 3.1.7 respectively.

4. Website shall pull information directly from the LTA API to ensure reliability and accuracy of data.

4.1. Website shall update the information every minute or whenever the respective icons are clicked to ensure constant accuracy of data.

4.1.1. Information includes bus arrival timings, taxi availability locations, carpark availability and traffic alerts.

Functional & Non-Functional Requirements

5. Website shall provide a search functionality.
    - 5.1. Website shall allow users to search for carparks by location or postal code.
    - 5.2. Website shall allow users to search for bus stops by name or code.
    - 5.3. Website shall allow users to view a list of traffic alerts.
    - 5.4. Website shall display “No results found. Please try again with a different search query” if no results are found.

6. Website shall provide AI-powered travel type recommendations.
    - 6.1. Website shall allow users to enter an end location into a search bar.
    - 6.2. When the user enters an end location, the website shall send the current location, end location, and relevant travel data(traffic, parking, taxi availability, bus availability) to the external AI model via an API call. The website shall then display the AI’s recommended travel option (see section 6.3) as the output to the user.
    - 6.3. Only 3 options are available:
        - 6.3.1. Car – if roads are clear and parking is available
        - 6.3.2. Taxi – if availability is high and roads are congested
        - 6.3.3. Bus – if taxi is too expensive and the place is easily accessible using bus.

7. Website shall have a user profile page
    - 7.1. In the profile page, the website shall display the user’s email and phone number.
    - 7.2. Website shall allow users to change their password.
    - 7.3. Website shall allow users to log out.

8. Website shall allow the user to click “Start Now” on the landing page to bring them to the general map view.
    - 8.1. Website shall allow the user to choose between different map views from the Navigation Bar.
    - 8.2. If user clicks any of the buttons, they are taken to a new screen consisting of a list menu and general map.
    - 8.3. Website shall allow user to choose Bus view.
        - 8.3.1. Website shall allow user to input a destination and the website shall display all bus stops along the computed route as icons with bus stop names and codes.
        - 8.3.2. User can also view total price, and estimated time of arrival based on the AI’s analysis.
    - 8.4. Website shall allow user to choose Taxi view;
        - 8.4.1. Website shall allow user to input a destination and view nearby taxis, current prices, and estimated time of arrival on the AI’s analysis.
    - 8.5. Website shall allow user to choose Drive view.

Functional & Non-Functional Requirements

8.5.1. Website shall allow user to input a destination and view relevant carparks nearby, estimated time of arrival, ERP charges, and relevant traffic alerts based on the AI's analysis.

8.6. All textual and numeric information is displayed in a tabular format beside the general map view.

Non-functional Requirements

Performance

- NFR1.1: The website must load within 3 seconds on a standard broadband connection (≥25 Mbps).
- NFR1.2: The system must return search results and LTA API data within 5 seconds. If the network or API is unavailable, the system must display a fallback message and allow cached or partial data access.

2. Usability

- NFR2.1: The interface must be intuitive and easy to use, with clear icons, consistent layouts, and descriptive labels.
- NFR2.2: Error messages (e.g., invalid login, no search results) must state the cause of the error in plain language and provide a specific next action.

3. Accessibility

- NFR3.1: The website must follow basic WCAG 2.1 standards for accessibility.
- NFR3.2: Users must be able to navigate all primary features (search, booking, account) using only the keyboard.

4. Reliability

- NFR4.1: The system must maintain 99.9% uptime to ensure reliability.
- NFR4.2: The data presented in the system must be updated at least once every 60 seconds to reflect the latest available information.

5. Scalability

- NFR5.1: The system must support up to 5,000 concurrent users without performance degradation.
- NFR5.2: The system must be scalable to accommodate new data sources, additional features, and up to 20,000 concurrent users in the future.

6. Security

Functional & Non-Functional Requirements

- NFR6.1: User data must be encrypted in storage and during transmission.
- NFR6.2: Password reset and account verification must be done through unique, single-use email links that expire within 15 minutes.

7. Compatibility

- NFR7.1: The website must work on the latest versions of Chrome, Firefox, Safari, and Edge.
- NFR7.2: The website must be responsive and adjust correctly to mobile, tablet, and desktop devices.

8. Maintainability

- NFR8.1: The system's code must be modular, documented, and follow standard naming conventions for easy updates.
- NFR8.2: Bug fixes and updates must be deployable with less than 5 minutes of downtime per release.